From Alert Fatigue to Augmented Defense: A Case for AI Copilots in Cybersecurity Operation Centers

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***Abstract*—Cybersecurity Operations Centers (CSOCs) act as one of the foremost tools for cyber defenses. However, security analysts often face overwhelming stress owing to excessive volumes of alerts, frequent false positives flagged by most existing tools, and growing mental strain. Such challenges could negatively impact threat detections by delays that cause slower incident response; thus, undermining overall security effectiveness. Common tools like Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) help automate tasks, but they often do not provide enough situational understanding, clear explanations, or real support for analysts making decisions. This paper presents a position on the role of copilots that are powered with AI, to strengthen Cybersecurity Operations Centers (CSOCs). We argue that copilots, integrating machine learning, large language models, and explainable AI (XAI), can (1) make analysts' work easier by grouping similar alerts and highlighting the most important ones first, (2) make threat detection and incident triage more accurate by adding useful background information, (3) build analyst trust and increase adoption by giving clear and understandable explanations, and (4) improve the quality of decisions by helping humans and AI work together effectively. These points are explained using logical arguments and insights from existing literature on interactive security operations i.e. human-in-the-loop security. This paper positions CSOCs as places where analysts work with AI support. It shows how copilots can assist human analysts to improve from just reacting to threats to being more proactive and flexible in their defenses. The paper also looks at what this means for research, everyday use, and guidelines to ensure AI copilots are used safely in CSOCs.**

Keywords— Security Operation Centers (SOCs), Cybersecurity Operations Centers (CSOCs), AI Copilot, Alert Fatigue, Threat Detection, Incident Response, Explainable AI (XAI), Human–AI Collaboration, SOAR

I. Introduction

Cybersecurity threats are becoming bigger, smarter, and more frequent. This puts strong pressure on organizations to protect themselves. CSOCs act as the main hubs where teams watch for attacks, detect them, and take action [1]. But even though CSOCs play a critical role, they face serious challenges. Analysts are overloaded with too many alerts, they deal with many false alarms, and they often feel mental fatigue. Together, these issues slow down threat detection and delay the response to real incidents [2].

To address these challenges, organizations have widely adopted Security Information and Event Management (SIEM) platforms and Security Orchestration, Automation, and Response (SOAR) solutions [3]. These tools collect alerts, automate some tasks, and follow set playbooks [3]. But they often do not give enough background, do not clearly explain why actions are suggested, and give little help to analysts when making important decisions [4].

This paper suggests that AI copilots are the next big step to make CSOCs better. They can use machine learning to find unusual activity, large language models to give background and explanations, and explainable AI to make results easy to understand. With these, copilots can change CSOCs from only reacting to problems into being proactive, flexible, and easier for analysts to use. We see AI copilots as an important way to improve CSOCs, and we shape our discussion around the following research questions:

RQ1: How can AI copilots make analysts' work easier by grouping alerts and showing the most important ones first?

RQ2: How can AI copilots improve accuracy and reduce false alarms when handling alerts?

RQ3: How do clear explanations from AI copilots help analysts trust and use them?

RQ4: How can analysts and AI copilots working together lead to better decisions?

II. Review of Literature

A. Challenges in Cybersecurity Operations Centers

Cybersecurity Operations Centers (CSOCs) are extremely important for providing a security shield for organizations. However, many reports show that analysts are overwhelmed by unsurmountable alerts every day[5]. Many of these alerts may be false alarms[6], which prove to be a waste of time and resources and may cause alert fatigue for the SOC analysts [5]. This makes it harder to quickly find real threats and respond in time[5][6].

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B. SIEM and SOAR Tools

As more and more organizations across the world face vicious cyber-attacks, it has been observed that traditional security measures have fallen short to provide them sufficient protection. Advanced security tools such as Security Information and Event Management (SIEM) systems and Security Orchestration, Automation, and Response (SOAR) have provided some better security mechanisms [7][8]. The SIEM tool analyses logs and creates intelligent security alerts based on threat detection programs embedded in it. SOAR has been programmed to guide analysts through workflows called "playbooks"; which help them through steps to manage and respond to security incidents [7]. These tools may prove to be useful, but they mostly depend on fixed rules and automation. They may be unable to show how an alert may be connected to a larger attack, and hence, they may not be able to explain why a specific action needs to be taken[8].

C. AI and Machine Learning in Security

Machine Learning is a subfield of Artificial Intelligence, which enables systems to learn from data, recognize patterns in data and make decisions without being specifically programmed for each situation. Machine learning can be adapted to cyber security systems to identify unusual patterns and thus possibly detect attacks. Machine learning models heavily depend on existing data sets, with Canadian Institute for Cybersecurity Intrusion Detection System (CICIDS) [9] and University of New South Wales – Network Based 2015 (UNSW-NB15) [10] being the most popular datasets for cyber security. Both datasets provide machine learning embedded cybersecurity tools to improve the capabilities for threat detection [9][10], however most machine learning based cybersecurity tools are standalone tools that do not entirely satisfy requirements for AI copilots as they are not fully capable to provide complete support to analysts at real time. Suck machine learning based tools are often termed as "black boxes" as they do not provide rationality for the decision suggested by them, thus casting some doubts over their reliability..

D. Explainable AI (XAI)

Explainable Artificial Intelligence (XAI) are specialized AI algorithms that are designed to provide explanations (which could be understood by humans) for each decision that they suggest. XAI systems use techniques such as SHAP (SHapley Additive exPlanations) [11] [12] and LIME (Local Interpretable Model-agnostic Explanations)[11], that provide explanations on the data that influenced the XAI system's decisions. Such explanations could provide to be a boon for cybersecurity analysts, by enabling them to understand the rationale behind a cyber threat alert generated and in turn provide them sufficient analysis to respond to the threat. XAI methods have been successfully applied to high risk fields such as healthcare [13] and finance [14]; however their practical applications to the field of cybersecurity especially in CSOCs are not as common..

E. Human–AI Collaboration

Rather than working disjointly, humans and AI systems may thrive if it is possible to enable them to collaborate to get tasks done collectively, i.e. Human-AI Collaboration. Human-AI collaboration has worked wonders in field such as healthcare [15] and aviation [16]. Some studies in cybersecurity have suggested that it is possible to mix analyst judgement with AI suggestions to enable them to detect threats more accurately and efficiently [17]. However, the study on AI copilots (Human - AI collaboration systems) is still in the infancy. Such systems, if designed appropriately, have the potential to achieve superior threat detection and prevention results as compared to analysts using tools for support only.

F. Gap in Research

Tools like SIEM, SOAR, and machine learning systems help with some CSOC problems, but they do not fully solve the issues of too many alerts, not enough useful information and unclear reasons for decisions. Research on Explainable AI (XAI) and human–AI teamwork shows promise, but these ideas are still not well connected in practice. At present, there is no clear framework that combines machine learning, large language models, and explainability into one system to directly support analysts.

As per the subsections covered above in Related Work section, we can see that SCOC tools have slowly but steadily progressed from using fixed rule based systems, to more smarter systems. Figure 1 below depicts this transition of CSOC technologies from alert collection (using SIEM) to task automation (using SOAR), to decision support systems (using Machine Learning) and then understandable logic (using XAI), and culminating into our suggested fully automated AI-Copilot systems. We can possibly and with some level of confidence, predict that the improvements in CSOC technologies could support cybersecurity analysts to implement robust threat detection and prevention strategies.

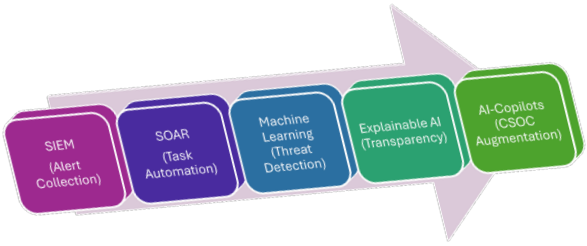


Fig. 1. **Evolution of CSOC tools toward AI copilots**

This paper aims to address this gap by suggesting AI copilots as a new way to make CSOCs more proactive, adaptive, and easier for analysts to use.

III. METHODOLOGICAL APPROACH

In our research paper, we have used a conceptual and qualitative research approach to present a particular position of using AI copilots in CSOCs. As our aim is to mainly present our position [18], we do not have the necessity of testing or validating any new or existing system. A position paper should point out what is missing in current knowledge, look at the facts, and then clearly state the author's opinion or recommendation [18]. Our main focus is to develop an argument strongly backed by literary and scholarly sources and present it in this research paper.

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We have identified three such main sources:-

1. Existing literature on CSOC or SOC operations, application of AI tools and techniques' in the field of cybersecurity

2. A thorough analysis of current SOC tools such as SIEM and SOAR to identify gaps in cybersecurity threat detection and related decision making.

3. Logical thinking about how AI copilots driven by machine learning and explainable AI could help to fill identified gaps.

Based on above sources and research objectives, four research questions (RQ1 to RQ4) we were developed, and have been presented in introduction section above

IV. Position and Core Arguments

As per the research objective, this paper presents the position that AI-copilots could possibly play an important role in improving Cybersecurity Operations Centers (CSOCs). Copilots most certainly carry an immense potential to act as intelligent assistants that work collaboratively with analysts rather than replace them. By creating an effective blend of existing and emerging technologies such as machine learning, large language models, and explainable AI; copilots could possibly help to reduce alert fatigue, to improve accuracy in threat detection and analysis, and support impactful decision making. The paper's position can be explained and corelated to the four research questions (RQ1 to RQ4) as seen below:-

A. Reducing Alert Fatigue and Cognitive Load (RQ1)

AI Copilots have the potential to make threat detection and related tasks much easier for CSOC analysts. AI copilots could classify alerts, group similar alerts together, try to apply filters to erase duplicate alerts and categorize alert risks. Such classification, grouping, suppressing duplicates and categorization of alerts as per risk quotient, could help CSOC analysts to focus on alerts that need full and undivided attention, rather than wasting valuable time on low risk alerts. This could also reduce the alert fatigue and increase speed to incident responses.

Other fields have successfully deployed AI Copilots; AI copilots have been extensively used in businesses and corporate houses to improve employee productivity by automating simple tasks such as sorting emails, creating smart reports, checking for errors, etc. [19]. In the field of education, AI copilots have been used successfully to support English language teachers through automation of tasks such as lesson plan creation, providing feedback on student submissions, explain grammar in simpler and more interactive manner [20]. This study emphasizes on application of these ideas to support CSOC analysts in field of cybersecurity defense..

B. Improving Accuracy and Reducing False Alarms (RQ2)

An important subfield of AI, Machine Learning (ML) thrives on algorithms that include a self-learning logic to enable such systems to learn from patterns in huge datasets. This fortifies performance of such systems without the need for re-programming them. ML can strengthen the ability of CSOC AI copilots to learn from patterns flagged in threats and take impactful decisions to avoid or thwart any serious cybersecurity threats [21]. Such ML fortified AI copilots for CSOC analysts also derive an unique capability of learning from past security incidents and hence develop an uncanny characteristic of reducing false alarms; an ability which humans may find extremely challenging to achieve without technological aid. This ability also provides a much wanted liberty to CSOC analysts to not waste valuable surveillance time and resources on trivial threats and instead focus on serious security threats and take timely and crucial decisions.

In fact, ML driven AI copilots have been used in other field quite successfully. In Neurophysiology, ML driven AI systems in collaboration with large language models (LLMs) have been used for analysis of brain signals and other medical data [22]. This helps surgeons to identify abnormal patterns in the patient data more accurately, thus enabling them to effectively reduce false alarms, especially by monitoring brain activity during critical surgeries. Once again this enables surgeons to devote all their invaluable time to medical procedure rather than responding to false alarms [22]. Similarly, in the field of Power Electronics, ML driven AI systems have been put to effective use to monitor devices such as inverters and converters [23]. These systems are quite proficient in detecting false alarms regarding device faults and with such fault tolerant characteristics, they have the tendency to reduce unnecessary shutdowns thereby improving overall stability of these systems [23].

C. Building Trust Through Clear Explanations (RQ3)

Explainable Artificial Intelligence (XAI) powered AI systems provide humans with capabilities to understand and interpret AI algorithms, thereby strengthening their trust on such systems [24]. Traditional AI systems are often termed as "black box" systems, as it is not very clear as to how they derive some conclusions or decisions. This certainly questions their trustworthiness. The "white box" XAI algorithms, when embedded into AI copilots, not only provide conclusions or decision suggestions similar to traditional AI systems; but they also provide deeper root cause analysis information that can boost users' confidence [24]. Thus, when XAI and ML powered AI copilot systems are created for CSOC analysts, they can share not only security alerts but also explain the cause of such alerts i.e. an alert was considered as critical as it matches a ransomware pattern. Such diagnostic and deep analytical information raises trustworthiness.

XAI based AI copilot systems have been used in different fields. In Control systems sector, AI system designers have used combination of deep learning and XAI techniques to detect faults and diagnose their root causes [25]. When used in manufacturing systems, such control engineering copilots can help to spot critical faults in sensors, and the collaborative XAI components can help to diagnose the fault, identifying which signals or measurements data flagged it. With such clearer information on the fault technicians may take crucial repair decisions in a timely manner [25]. In the software engineering field, XAI powered AI copilots have been successfully designed to assist software coders to not only create code but also test it effectively. Recent tools such as GitHub Copilots have been developed with abilities to not only suggest lines of code for different software modules, but also to provide explanations about the suitability and applicability of the suggested lines of code [26]. It is natural human instinct that they trust things that they more clearly understand. The code's understandability and applicability provided by the GitHub's AI copilot enriches developer trust in the copilots and thus increases their confidence in such tools.

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D. Improving Decision Quality Through Collaboration (RQ4)

The most valuable deliverable of AI copilots is to provide a holistic ecosystem for humans and AI copilots to work together collaboratively on common agenda. The objective of creating AI copilots is not to replace human experts but rather to support human experts by suggesting solutions or decisions based on observed patterns and providing understandable explanation for such decisions. Such an arrangement creates a formidable techno-human team, each member bringing a unique value; humans bringing rich field experience and technical expertise; and technology bringing speed, reasoning, and data crunching abilities.

Many fields have successfully implemented this techno-human teamwork using AI copilots. In Human Resources (HR) field, AI copilots have been successful in providing personalized experiences to employees [27]. HR officers (HROs) use such AI copilots to analyze feedback, synthesize survey data, and create strategies to improve workplace satisfaction [27]. One such tool Microsoft Copilot has been popularly used for HR related decision making [27]. AI copilots also find their relevance and applicability in the field of academics. Some researchers have studied use of Microsoft Copilot tool in a costume designing academic program [28], thereby enriching creative education. This study concluded that Microsoft Copilot was successful in helping costume designing course students to explore newer design ideas, create creative designs, and understand the designing process more comprehensively [28].

The above successful examples give us some confidence to state that AI Copilots may have a potential use case in CSOC operations. CSOC analysts could work collaboratively with AI copilot to provide a robust cybersecurity defense. The AI copilot could analyze data in a speedy manner to analyze patterns, compare the patterns with known threats and suggest possible actions on threat prevention and other related responses. This teamwork can use the expertise of CSOC experts along with the advanced algorithms of the AI copilot (reinforced by technologies such as ML and XAI) to provide formidable security defense mechanisms for organizations.

V. Discussion and Implications

After reviewing the above sections I. Introduction, II. Related Work, III. Methodological Approach and IV. Position and Core Arguments, we can say that AI copilots have a solid case to be used in CSOCs to support CSOC analysts to become more alert, accurate and decisive. This paper has explained how AI copilots can help CSOC analysts to transition from condition of severe alert fatigue to a stronger position to provide augmented cybersecurity defense. The findings presented in these research paper segments demonstrate a possible proactive role that AI copilots can play to support CSOC analysts to improve CSOC security management. This proactive role of AI copilots support CSOC analysts to overcome alert fatigue and take watchful security decisions.

The discussion and implications section now discusses some implications of using AI copilots for CSOC centers; focusing on three areas of concern - implications on research, implications on practice, and implications on governance and ethics.

A. Implications on research

Our research has reviewed existing research and presented our findings to demonstrate the usefulness of the idea of using AI copilot in CSOC operations. However, more research may be necessary to investigate possibilities of grouping alerts based on parameters, use of XAI methods to create explainable and understandable suggestions, and other ways the AI copilot could support CSOC analysts. Research could possibly include small case studies to test the idea or a prototype with imaginary scenarios and data, validating its alertness behavior and suggestions. Research may also check the performance ot the AI copilot by feeding it false positives and observing its reaction. Research could also possibly involve collaborative use of machine learning and large language models (LLMs) to enable AI copilots to provide clear and reliable solutions/ suggestions/ reactions to perceived threats.

B. Implications for practice

This section discusses the implications of AI copilot for actual practice at CSOC centers. There are some AI copilots which are actually in practice for CSOC centers, and such successful implementations of AI copilots have managed to bring a change in the way CSOC analysts work. Existing security copilot tools such as Microsoft Security Copilot, Torq Socrates, Gurucul AI-SOC Analyst and Darktrace Cyber AI Analyst show that AI copilots can help to sort alerts, to check incidents, and to take reactive action. Microsoft Security Copilot has successfully helped CSOC analysts to investigate and respond to security alerts by using large language models (LLMs) [29]. Torq Socrates AI tool specializes in automating SOC workflows [30]. Gurucul organization's AI-SOC Analyst tool has a unique algorithm that scores security risks, classifies security alerts, and creates action recommendations at real time [31]. Darktrace Cyber AI Analyst specializes in machine learning and expert systems and used this mastery to investigate security threats thereby producing human understandable summaries of incidents for CSOC analysts [32]. However, for these wonderful tools to succeed, security organizations should provide adequate training to CSOC analysts to work collaboratively with AI copilot tools. They should not forget that main principle of human-machine collaboration; that the human should always stay in control and take key security-related decisions with help of AI copilot assistants. Most of the above tools use some of the technologies that we have discussed in our paper, i.e. SIEM, SOAR, ML and XAI; either completely or partially. Thus, this supports the central argument of our paper, that AI copilots could successfully be implemented to augment CSOC operations. Hence, the discussion on above implementations shows that AI copilots are feasible for use as augmentative technologies in CSOCs and they can possibly reduce alert fatigue in CSOC analysts and provide them with explainable suggestions to take security decisions in proactive manner.

C. Implications on research

At present and possibly more so in the future, AI copilots are likely to become part of human-AI collaborative team for CSOC operations; it is necessary for organizations to make policies to ensure their ethical use. In order to eliminate future legal issues, it would be vital to write clear policy rules regarding who would be at fault, in case the AI copilot makes a wrong suggestion that could potentially harm the security defense. Careful scrutinization should be conducted so that the AI algorithms do not create unfair bias and maintain the

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privacy and security of data. This means that every organization that employs AI copilots for CSOC operations should apply clear and strong IT and security governance policies.

VI. CONCLUSION

This position paper presented the researchers' point of view about the role of AI copilots in enabling a technology driven transition in Cybersecurity Operations Centers (CSOCs) from reactive decision making to proactive and analyst friendly intelligent hubs. The research team has suggested the use of time-tested technologies such as SIEM and SOAR with emerging AI technologies such as ML and XAI; to help reduce alert fatigue, to improve accuracy in security decision making, to classify threats (based on risk parameters) so as to focus only on critical issues rather than trivial ones. The paper has argued strongly that above technologies can boost the CSOC's augmented defense capabilities.

This paper has also discussed how the AI copilot tools in use for CSOC operations - Microsoft Security Copilot, Gurucul AI-SOC Analyst, Torq Socrates and Darktrace Cyber AI Analyst; use the same technological vision (SIEM, SOAR, ML, XAI) that we presented as our position in this paper. Hence, this provides a strong real world applicability to our position that AI copilots can positively augment CSOC operations by providing the multiple benefits that were already discussed. Our position has maintained focus on Human-AI collaboration, with the primary role for CSOC analyst and a supportive role for the AI copilot.

The paper also showed that many tools—such as Microsoft Security Copilot, SOC Copilot (OpenAI Custom GPT), Gurucul AI-SOC Analyst, Torq Socrates, Darktrace Cyber AI Analyst, Prophet Security, and GitHub Copilot—already use parts of this vision. This gives strong real-world support to the position that AI copilots are a practical and achievable step in the evolution of modern security operations. This collaborative endeavor is not merely for automation of CSOC tasks, but rather a trustworthy association that has the potential to thwart most given cybersecurity threats at real time and in a proactive manner. The paper has presented not only the technical aspects but also research aspects and ethical aspects, thus demonstrating a need for an ethical AI approach to ensure log term success.

The paper has also discussed AI copilots in other fields such as education sector, corporate sector, healthcare sector, power electronics sector etc. Such diverse examples have demonstrated that Human-AI collaborative efforts are much needed in supporting humans in complex decision making. Lessons learnt from cross-disciplinary implementations provides strong insights into our system of interest i.e. CSOC operations. It also provides a level of confidence in our vision for this paper.

It would be unwise to state that the paper doesn't have any limitations. Like any other research work, this position paper has its own limitations. As this is a position paper, it has not included any real prototype testing or primary data analysis. This paper has relied on strong review of existing literature and industry uses cases. In future, this research may be extended to include experimental studies, user evaluations, or prototype testing to measure how well AI copilots perform in real SOC environments.

As a concluding remark, we state that AI copilots could have immense potential to improve CSOC operations. The AI copilots, if used with the right blend of technology, human control, and ethical governance, have a true potential to help CSOCs to move from reactive defense to proactive, adaptive, and trustworthy cybersecurity.

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